



Project Assignment #1

Answer the following questions regarding Stata and your ACS data set. To complete this assignment, you will upload three things to canvas: (1) a Word or PDF document containing your complete answers to the questions below; (2) your Stata “do” program; and (3) your Stata “log” file. The Stata “do” program (and therefore your “log” file) must include comments stating which question you are answering with the code you write.

The first two questions do not need to appear in your Stata do and log files.

1. In your project pre-assignment, you created an extract from the ACS data. This extract contains several variables for working individuals aged 25 to 59. For this project assignment, you will load this data set into Stata and limit the sample to a specific occupation that you will choose. The first step in this process is to choose an occupation that you would like to study.
 - a. Go to the IPUMS-USA website, click on “variables”, then on “person,” and then on “work.” Click on the “codes” link for the variable “OCC” (occupation). Click on the link for the “ACS/PRCS Occupation Codes”. This page contains a list of the occupation codes used in the ACS data. As you can see, there are a lot of occupations to choose from. Look through the list and choose an occupation (or two or three related occupations) that you would like to study. We will be studying wages and returns to education within the occupation that you choose. Write down the “ACS OCC code” for the occupations that you are interested in studying. You can change your occupation at a later time. For example, I am interested in studying the wages of elementary and high school teachers, and so I would write down:

2310 - Elementary and Middle School Teachers
2320 - Secondary School Teachers

Note: most occupation codes did not change after 2000, but some did. There is also a new occupation codebook starting in 2018. If you choose an occupation that does change codes, you will need to make note of this and modify the “keep” command in part (d) below.

These instructions assume that you are using Stata in a computer lab where you where you may be working on different computers on different days. If you are only using Stata on your own computer, you do not need to use a USB drive and can store your work on your computer or cloud drive.

- b. Insert your USB drive and open Stata. In Stata, click on “File” and then on “Change Working directory...” Change the working directory to the folder on your

USB drive that contains your data. In the Stata “Command” window, type “dir” and press enter. You should see your “.dat” file and any other files you downloaded.

- c. Open the Stata do file editor. You can do this by clicking on the “do file editor icon” at the top of the Stata window, by clicking on Window – Do file editor, or by pressing Ctrl-9. Paste the following Stata code onto your do file editor and then save the do file in the same director as your ACS (.dta) data.

```
*****
*** econ5813_IPUMS_EXTRACT.do ***
*** ----- ***
*** Created by {your name} ***
*****

*****
*** This program creates variables and estimates ***
*** regressions for ECON 5813 Project Assignment #1 ***
*****

*****
*** Begin the log file ***
*****

capture log close
log using econ5813_IPUMS_EXTRACT.log, replace

set more off
set linesize 255
set varabbrev off

*****
*** Load the ACS data (2009 - 2022) ***
*** ----- ***
*** Downloaded from https://usa.ipums.org/usa ***
*****

clear all
use ./usa_00102.dta
describe, short

*****
*** Check that the data is correct ***
*** ----- ***
*** Modify this to be consistent with your data. ***
*** Add additional data integrity checks as you ***
*** think of them. ***
*****

* Data includes years 2009 through 2022
tab year, missing
assert inrange(year,2009,2022)==1 & r(r)==14
```

```

* Data includes individuals between the ages of 25 and 59
    tab age, missing
    assert inrange(age,25,59)==1

* Data includes individuals who are employed
    tab empstat, missing
    assert empstat==1

*****
*** Limit the sample to my occupation ***
*****

    keep if occ=={your 1st occ code} | occ=={your 2nd occ code}
    describe, short

*****
*** Save the Data ***
*****

save econ5813_IPUMS_EXTRACT.dta, replace

```

- d. Change the partes in the in the code above that are highlighted in yellow to be consistent with your data. For example, replace {your 1st occ code} with the first occupation code that your selected in step (a) above. Do the same for {your 2nd occ code}. For example, my code (for teachers) will look like:

```
keep if occ==2310 | occ==2320
```

If you selected one occupation, then delete everything on this line after the first occupation code (keep if occ==2310). If you selected more than two occupation, you can add more occupation codes to this line:

```
keep if occ==2310 | occ==2320 | occ==2330
```

- e. Save the changes to the “.do” program by clicking the save button. Run this program by clicking Tools, then Execute (do). You can also run this program by clicking on the “Execute (do)” button at the top of the Stata Do-file Editor window. It may take a few minutes for the Stata program to run.
- f. Click back on the Stata main window to see the output of the Stata “.do” program. When the program is finished, you will see “end of do-file” printed on the screen. If you see an error message (printed in red), then you have a typo in your code. Fix this problem before continuing on. If the program runs correctly, you will see the output from the two “describe, short” commands that you added to the end of the program. You should see that the number of observations in your data decreased after the “keep” command. You need to have more than 500 observations in your data set. If you do not, then you selected an occupation with too few workers. If this is the case, then you need to go back to step (a) and select an entirely new occupation

or add additional occupations to your data. Do not proceed to the next step until you have a data set with at least 500 observations.

Read the “Getting Started” section in the Stata guide found here:

<https://www.ssc.wisc.edu/statistics/stata/>

Read all sections in the “Getting Stated” section before answering the questions below. You may not understand everything in the guide the first time you read it. This is ok. After you complete this Project Assignment, review the “Getting Stated” section again. It would be a very good idea to read “Data Wrangling” section as well.

Use the Stata guide and Stata’s “help” command to familiarize yourself with the following Stata commands (i.e., type “help *command*” in Stata for a description of each command):

log	cd	use	list
summarize (sum)	tabulate (tab)	tabsum	drop, keep
rename	generate (gen)	replace	describe (d,s)
assert	save	clear	egen
label	by	ttest	regress
inlist	Inrange	r()	collapse

2. Explain what the following commands will accomplish in Stata. Include in your explanation the definition of each variable (i.e., don’t say “for sex equals one”, say “for females”).

- a. `tabulate educd`
- b. `tabulate educd, nola`
- c. `summarize incwage if educd>=10`
- d. `summarize incwage if educd <10 & sex==1`
- e. `drop sex incwage`
- f. `drop if sex==2`
- g. `keep sex incwage`
- h. `list sex incwage`
- i. `tabulate school sex`
- j. `tabulate school sex, sum(incwage)`
- k. `generate diploma = 1 if educd>=63`
- l. `replace diploma = 0 if educd<63`
- m. `assert inlist(sex,1,2)==1`
- n. `assert inrange(educd,2,116)==1`

- o. `tabulate statefip, missing`
`assert statefip>=1 & statefip<=56`
`assert r(r)==51`

- p. `generate year16 = year-age+16`
`label variable race_cat5 "Year individual turned 17"`

```

q.    preserve
      collapse (mean) age educd ftotinc, by(statefip year)
      restore

r.    sum incwage
      return list
      display r(sum)/r(N)

```

Note: You do not need to include the Stata commands above in your do and log files, but you should run each of these commands interactively with your data to verify that the command does what you think it does.

From this point on, you can use Stata's interactive mode to test commands; however, all of your final work should be done in one Stata do program. Call the do program "econ5813pa1.do" and the log file "econ5813pa1.log." Label every variable that you create in your do program, even if the question does not ask you too.

Begin writing a new Stata do program by following these steps (these steps can be done interactively in Stata, and do not need to appear in your Stata do and log files):

- Step 1. Start Stata. What is Stata's working directory (use "pwd")?
- Step 2. Change the working directory to the folder on your USB drive.
- Step 3. List all of the files in your working folder (use "dir").
- Step 4. Open Stata's do file editor.
- Step 5. Write your Stata program, save it, run it, repeat.

Your Stata program "econ5813pa1.do" should start with the data you created named "econ5813_IPUMS_EXTRACT.dta." The program should include the same program headers (slightly edited) and include all the data integrity checks.

3. Load your data into memory using the Stata command "use econ5813_acs_d1.dta".
 - a. Describe the data. How many variables are there? What is the size of the data set? Do not "list" or print out every observation in your dataset. Type the command "compress" and describe the data set again. What changed and why?
 - b. List the variables age, sex, and state for the first 20 observations only.
 - c. Sort the data by state and age.
 - d. Once again, list the variable age, sex, and state for the first 20 observations. Are the 20 listed observations in part (b) different from those in part (d)? If so, why? If not, why not?
 - e. List the variables age, sex, state and incwage for the first 10 female observations and for the first 10 male observations. When you are done, you should have listed exactly 20 observations. (Note: this task is not as straightforward as it may sound. There are many ways to do this. Once you have accomplished it, think of other ways you might do the same thing).

4. Clear the data from memory, and then load your data back into memory.

- a. Eliminate the variable “empstatd” from the data set.
- b. Create the variable “weeksworked” using the following commands:

```
generate weeksworked = .
replace weeksworked = 7 if wkswork2==1
replace weeksworked = 20 if wkswork2==2
replace weeksworked = 33 if wkswork2==3
replace weeksworked = 43.5 if wkswork2==4
replace weeksworked = 48.5 if wkswork2==5
replace weeksworked = 51 if wkswork2==6
label variable weeksworked "Weeks worked"

tab weeksworked, missing
assert inlist(weeksworked, 7, 20, 33, 43.5, 48.5, 51) == 1 & r(r) == 6
```

- c. Eliminate from your data set anyone who works 26 weeks or less out of the year.
 - d. Eliminate anyone from your data set who works less than 20 hours per week or more than 60 hours per week.
 - e. How many observations remain in you data set?
 - f. Create a new variable called “female” that is equal to “1” if the person is female and “0” if the person is male. Create a similar dummy variable named “male”.
 - h. Verify that the dummy variables you created are mutually exclusive and exhaustive by using the command: “tab female male, missing”.
 - i. How many of your observations are men and how many are women?
5. Start with all the variables you created in #4.
- a. What occupations does your sample include? (Use “tab occ” to answer this question). Are those the occupations that you intended to include?
 - b. Create a new variable called “wage” that is equal to a person’s yearly wage or salary income divided by the total number of hours a person works in a year. Hint: your will have to use three variables: incwage, uhrswork, and weeksworked.
 - c. What is the average wage of all of the people in your data set?
 - d. What is the average wage of people in your data set between the ages 20 and 30?
6. Continue on with all the variables you created in #4.
- a. What is the average wage of all men in your data set? What is the average wage of all women in your data set?
 - b. Can you reject the hypothesis that the true average wage difference between men and women is zero, assuming equal variance? Use the 90% confidence level (Compute this calculation by hand using the t-stat formula in Problem Set #2, Question 9, and then verify your results using Stata “ttest” command).

- c. Can you reject the hypothesis that the true average wage difference between men and women is 5, assuming equal variance? Use the 90% confidence level (Compute this calculation by hand using the t-stat formula in Problem Set #2, Question 8, and then verify your results using Stata “ttest” command).

Note: For parts (b) and (c), “by hand” means on paper. However, if you want to calculate this t-stat in Stata using the saved results from “summarize” commands and either local macros or scalar variables, that would be fine too.

7. Continue on with all the variables that you created in #6:

- a. Create the variable “schoolyr” using the following commands:

```
gen float schoolyr = 0
replace schoolyr = 2.5 if educd<=17
replace schoolyr = 5.5 if educd>=20 & educd<=23
replace schoolyr = 7.5 if educd>=24 & educd<=26
replace schoolyr = 9 if educd==30
replace schoolyr = 10 if educd==40
replace schoolyr = 11 if educd==50
replace schoolyr = 12 if educd>=60 & educd<=64
replace schoolyr = 13 if educd>=65 & educd<=71
replace schoolyr = 14 if educd>=80 & educd<=90
replace schoolyr = 16 if educd>=100 & educd<=101
replace schoolyr = 18 if educd>=110 & educd<=115
replace schoolyr = 20 if educd==116
```

Label the variable `schoolyr` as “Years of schooling”. Tabulate the variable `schoolyr` with the “missing option. After your tabulation, include the command:

```
assert inrange(schoolyr,2.5,20)==1 & r(r)==12
```

- b. Estimate the regression: $wage_i = \alpha + \beta schoolyr_i + \varepsilon_i$
- i. According to this regression, what is the association between education wages?
- c. Estimate the regression: $wage_i = \alpha + \beta_1 schoolyr_i + \beta_2 age_i + \varepsilon_i$
- i. According to this regression, what is the association between education wages? How does this answer compare to the answer to part (b)? Is this what you would have expected?
- d. Estimate the regression (you may do these calculations by hand):

$$wage_i = \alpha + \beta_1 schoolyr_i + \beta_2 schoolyr_i^2 + \beta_3 age_i + \beta_4 age_i^2 + \varepsilon_i$$

- i. What is the marginal effect of education on wages?
 - ii. What is the marginal effect of age on wages?
- e. Repeat part (d) separately for men and women.
 - i. Is the marginal effect of education the same for men and women?
 - ii. Is the marginal effect of age the same for men and women?

(Note: you don't need to answer parts (i) and (ii) with formal hypothesis tests. Simply comment on how the marginal effects of age and education compare between men and women).

- f. Estimate the regression:

$$wage_i = \alpha + \beta_1(schoolyr_i - 12) + \beta_2 \ln(age_i - 20) + \beta_3(schoolyr_i - 12) \times \ln(age_i - 20) + \varepsilon_i,$$

- i. What is the estimated (i) marginal effect and (ii) elasticity of age on wage for someone with the average education, age, and wage? Interpret each number using simple non-technical words.
 - ii. Using simple non-technical words, what is the economic interpretation of the OLS estimate of β_2 ? Refer to the actual number.
 - iii. Using simple non-technical words, what is the economic interpretation of the OLS estimate of the constant in this regression? Refer to the actual number.
8. Estimate the regression: $wage_i = \alpha + \beta_1 schoolyr_i + \beta_2 age_i + \varepsilon_i$, using “robust” standard errors. We have not discussed robust standard errors yet, so just use the “regress” command with the “robust” option, and think of the “robust” standard errors in the same way you would think of regular standard errors.

After you estimate this regression, execute the command “generate PersonID = _n” and then the command “expand 2”. Explain what these two commands did to your data.

Next, estimate the same regression again (including the “robust” option). Did $\hat{\beta}_1$ and $\hat{\beta}_2$ and $SE_{\hat{\beta}_1}$ and $SE_{\hat{\beta}_2}$ increase, decrease, or stay the same? Explain why (you do not need to prove anything, just explain using logic and/or intuition)? Explain why you would never use the output from this second regression when doing research.

Finally, estimate the regression a third time, replacing the “robust” option with “vce(cluster PersonID)”. You do not need to explain why, but how does $\hat{\beta}_1$ and $\hat{\beta}_2$ and $SE_{\hat{\beta}_1}$ and $SE_{\hat{\beta}_2}$ from this regression compare to the previous two? Explaining why will be the subject of future lectures in Econometrics I & II.

9. The book Mastering 'Metrics: The Path from Cause to Effect by Angrist and Pischke, presents the intuition of the multiple regression model in a slightly different way (starting on page 89). Some find this intuition enlightening, while others find it confusing. Below is an example of deriving the OLS coefficients through “residual regressions”. Using your ACS data, estimate the following regressions:

$$(1) \quad \text{schoolyr}_i = \alpha + \beta_1 \text{age}_i + \beta_2 \text{female}_i + \varepsilon_i.$$

$$(2) \quad \text{wage}_i = \alpha + \beta_1 \text{age}_i + \beta_2 \text{female}_i + \varepsilon_i.$$

Create two variables called “*schoolyr_resid*” and “*wage_resid*” equal to $\hat{\varepsilon}_i$ for each regression. To do this, use Stata’s “predict” command. Verify that the average $\hat{\varepsilon}_i$ for each regression is zero. Then estimate the following regressions:

$$(3) \quad \text{wage_resid}_i = \alpha + \beta_3 \text{schoolyr_resid}_i + \varepsilon_i.$$

$$(4) \quad \text{wage}_i = \alpha + \beta_4 \text{schoolyr}_i + \beta_5 \text{age}_i + \beta_6 \text{female}_i + \varepsilon_i.$$

What is the relationship between $\hat{\beta}_3$ and $\hat{\beta}_4$ in your Stata output? Explain why this relationship exists. You do not need to derive a mathematical proof. You should give an intuitive explanation using terms like “explained variation / unexplained variation” and/or “correlated / uncorrelated”.

Note: The Mastering ‘Metrics book describes the “residual regression” as:

$$(5) \quad \text{wage}_i = \alpha + \beta_7 \text{schoolyr_resid}_i + \varepsilon_i.$$

If you estimate regression (5), you will find that it is identical to regression (3), except that the standard errors are wrong. Nevertheless, you may find the interpretation of β_7 to be more straightforward than the interpretation of β_3 .